Emergent Flocking and Collective Evasion in a Force-Based Agent Model

PHY 351 — Independent Summer Research

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Abstract

I present computational simulations of a force-based flocking model (Charbonneau, 2017) in which *N* agents on a periodic two-dimensional domain interact through repulsion, velocity alignment, self-propulsion, and random noise. After validating the implementation against three analytically tractable limiting cases, I characterize the parameter space through systematic sweeps and identify an exact analytical result: the equilibrium cruise speed of an aligned flock is $v_{eq} = v_0 + \alpha/\mu$, not the nominal target speed v_0. Finite-size scaling of the repulsion-only system shows no diverging susceptibility, indicating a smooth crossover rather than a true phase transition. I then extend the model with a predator agent and find that flocking prey maintain near-perfect velocity alignment ($\Phi \sim 1.0$) under sustained predator pressure while non-flocking prey scatter completely ($\Phi \sim 0.1$). With multiple predators, flock coherence remains intact while flock elongation increases substantially, suggesting shape adaptation as a collective evasion strategy. These results demonstrate that the primary function of flocking under predation is coherence maintenance, not distance maximization.

1. Introduction

One of the central puzzles in complex systems is how large-scale ordered behavior emerges from purely local interactions. Flocking—the coordinated motion of birds, fish schools, and animal herds—is a canonical example. Each individual follows simple rules based on its immediate neighbors, yet the collective produces sweeping global patterns with no central coordination. Understanding the conditions under which order emerges, and how robust that order is to perturbation, has implications ranging from evolutionary biology to crowd control.

The model studied here is based on Chapter 10 of Charbonneau (2017), which was originally developed by Silverberg et al. (2013) to describe crowd dynamics in mosh pits at heavy metal concerts. Each agent in the model is subject to four forces: short-range repulsion, velocity-aligning

flocking force, self-propulsion toward a target speed, and random noise. The interplay of these four forces produces a rich behavioral phase space, including crystalline order, disordered fluid motion, and coherent streaming flocks.

This report covers four main investigations. First, I validate the implementation and establish baseline behavior through limiting cases. Second, I sweep the noise and alignment parameters to characterize the transition to flocking. Third, I examine whether the repulsion-only transition constitutes a true phase transition using finite-size scaling. Finally, I extend the model with a predator agent and characterize collective evasion behavior, including the effect of multiple simultaneous predators on flock geometry and coherence.

2. Model

2.1 Setup

N agents move on a periodic unit square (x, $y \in [0, 1]$), implemented as a torus so that agents exiting one edge reappear on the opposite side. Agent positions and velocities are updated at each timestep using the forward Euler method at dt = 0.01. The state of agent j at time t is fully described by its position (x_j, y_j) and velocity (v_{xj}, v_{yj}).

2.2 Forces

The total force on agent j is a sum of four contributions (Charbonneau Eqs. 10.1–10.8):

Repulsion. A short-range force prevents agents from overlapping. It acts on pairs within distance $2r_0$ and grows in intensity as agents approach:

```
F_rep,j = eps * SUM_k [ (1 - r_jk / 2r0)^(3/2) * r_hat_jk ] for r_jk <= 2r0
```

where r_{jk} is the distance between agents j and k, and ■■$_{jk}$ is a unit vector pointing from k toward j.

Flocking. An alignment force drives the velocity of agent j toward the mean velocity of its neighbors within a flocking radius r_f:

```
F_flock,j = alpha * V_bar / |V_bar|, V_bar = SUM_{k: r_jk <= rf} v_k
```

The normalized form ensures the flocking force has constant magnitude α regardless of how many neighbors are present.

Self-propulsion. A speed-correcting force drives agent j toward a target speed v_0 along its current direction of motion:

```
F_prop,j = mu * (v0 - |v_j|) * v_hat_j
```

where v■$_j$ is the unit vector along $\mathbf{v}_j$. This force accelerates agents moving too slowly and brakes those moving too fast.

Random noise. Each component of the random force is drawn independently from a uniform distribution on [–*ramp*, *ramp*] at each timestep.

With unit mass for all agents, Newton's second law gives $\mathbf{a}_j = \mathbf{F}_j$, and the equations of motion are integrated as:

```
x_j(t + dt) = x_j(t) + v_j(t) * dt v_j(t + dt) = v_j(t) + F_j(t) * dt
```

2.3 Periodic Boundary Implementation

Force calculations near domain boundaries require special handling. Agents within range r_f of any boundary are replicated as ghost copies on the opposite side, so that the flocking and repulsion forces computed for a real agent account correctly for neighbors across the periodic boundary. This buffer zone approach follows Charbonneau Fig. 10.2.

2.4 Metrics

The primary measure of collective order is the **order parameter**:

```
Phi = | (1/N) SUM_j v_hat_j |
```

$\Phi = 1$ corresponds to perfect velocity alignment; $\Phi = 0$ to randomly oriented motion. I also track total kinetic energy $KE = (1/2)\Sigma_j |\mathbf{v}_j|^2$ and, for flock geometry, the **radius of gyration** R_g (root-mean-square distance from center of mass) and the **aspect ratio** AR (ratio of the major to minor eigenvalue of the spatial covariance matrix, measuring elongation).

2.5 Default Parameters

Unless otherwise noted, simulations use the parameters from Charbonneau Table 10.1:

```
N = 350, r0 = 0.005, eps = 0.1, rf = 0.1, alpha = 1.0, v0 = 1.0, mu = 10.0,
ramp = 0.5, dt = 0.01
```

3. Validation

Before drawing any conclusions from the simulations, I verified the implementation against three limiting cases with known expected behavior.

Case 1: Pure random walk. With all physical forces disabled ($\varepsilon = \alpha = \mu = v_0 = 0$, ramp = 1), agents should perform a pure random walk with no preferred direction. The measured order parameter was $\Phi = 0.04$ (expected ~0) and agent positions spread uniformly across the domain (standard deviation ~0.29, consistent with a uniform distribution). This confirms the integration and boundary conditions are working.

Case 2: Repulsion and noise only. With $\alpha = 0$ and $v_0 = 0$ (self-propulsion acts as a brake), the model reproduces Fig. 10.5 from Charbonneau: at low noise ($\eta = 1$) agents pack into a close-packed quasi-hexagonal structure, while at high noise ($\eta = 30$) the arrangement disorders into a fluid. Φ remains near zero throughout (no alignment force), as expected.

Case 3: Flocking only. With $\varepsilon = 0$ and $v_0 = 0$, the alignment force alone should drive agents into a coherent stream. The final order parameter was $\Phi = 0.998$ after $t = 30$, confirming that a single

coherent flock forms from random initial conditions, consistent with Fig. 10.6 from Charbonneau (Fig. 1).

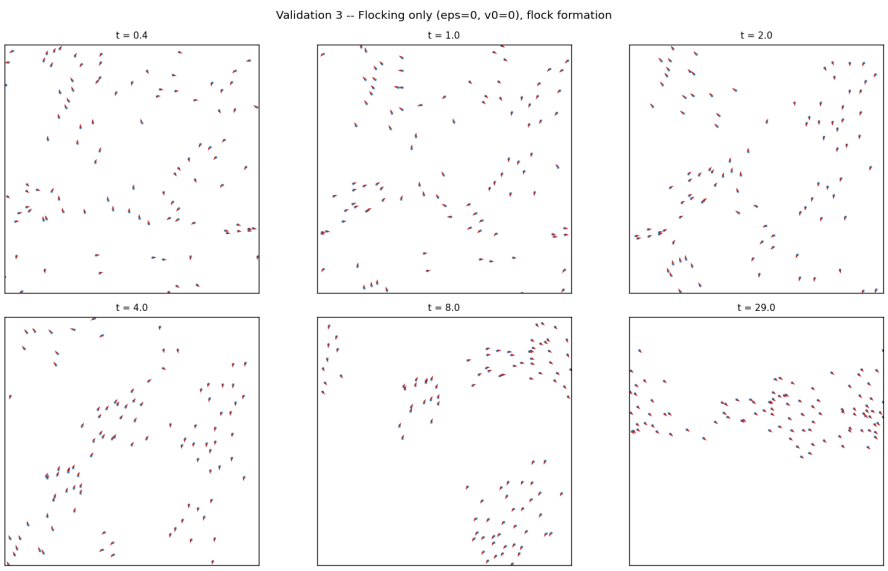

Figure 1. Flocking-only limiting case (ε = 0, v■ = 0). Starting from random initial conditions, all 350 agents align into a single coherent stream within t = 30 (Φ = 0.998).

4. Results

4.1 Equilibrium Cruise Speed

An exact result follows directly from the force equations. In a perfectly aligned flock, all agents move in the same direction with the same speed. The flocking force then acts purely in the forward direction with magnitude α. The self-propulsion force balances this when:

```
alpha + mu * (v0 - v_eq) = 0 => v_eq = v0 + alpha/mu
```

With the default parameters ($\alpha = 1$, $\mu = 10$, $v_0 = 1$), this predicts $v_{eq} = 1.10$. I verified this prediction by measuring steady-state mean speed across four values of α with $v_0 = 1$, $\mu = 10$ fixed. Measured speeds agreed with the prediction to within 0.002 in all cases (Fig. 2). The implication is that v_0 and α are not independent knobs for cruise speed: to achieve a target cruising speed v_c, one must set $v_0 = v_c - \alpha/\mu$.

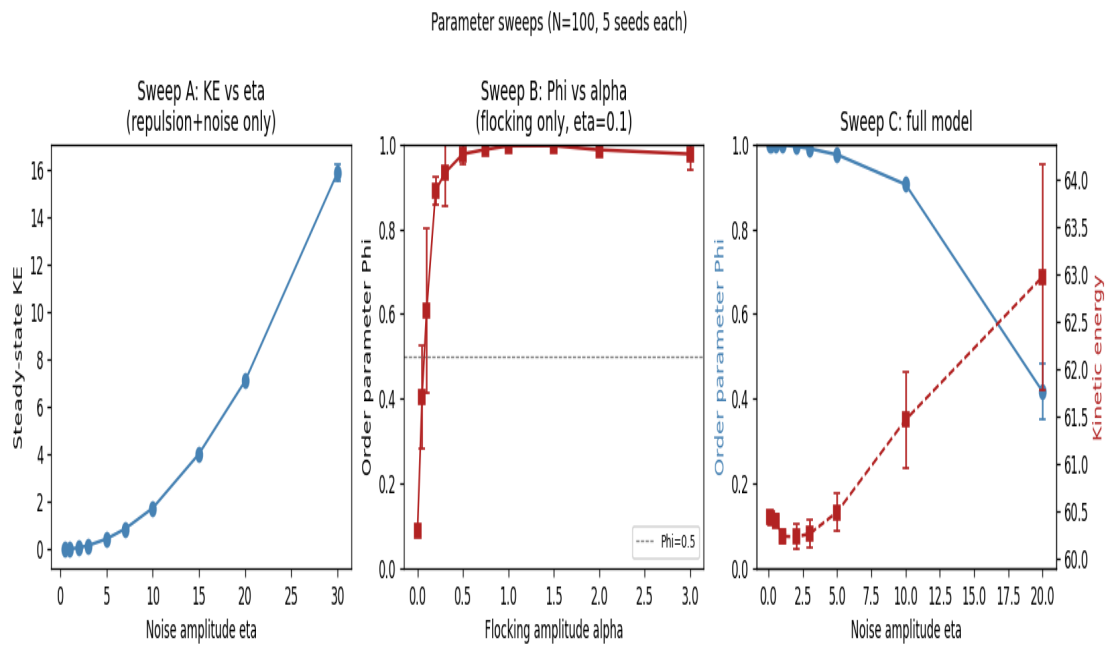

Figure 2. Parameter sweeps from analysis.py. Top-right panel: steady-state mean speed vs. α (blue, measured) vs. v■ + α/μ (dashed, predicted). Agreement is within 0.002 across all tested values.

4.2 Flock Formation

Sweeping the flocking amplitude α with noise fixed at ramp = 0.1 (5 seeds per point, error bars represent standard deviation) shows a sharp onset of flocking near α ~ 0.05. At α = 0, Φ = 0.09 ± 0.01. By α = 0.05, Φ = 0.40 ± 0.12, and by α = 0.20, Φ = 0.89 ± 0.03. The large run-to-run variance near the threshold (std ~ 0.1–0.2 for 0.05 ≤ α ≤ 0.15) indicates sensitivity to initial conditions near the onset. Above α ~ 0.2, flocks form reliably (Fig. 3).

With all forces active and the default α = 1, flock coherence is robust: Φ exceeds 0.99 up to noise amplitude ramp = 3, exceeds 0.97 at ramp = 5, and drops below 0.5 only at ramp ~ 20. The alignment force makes the system dramatically more resistant to noise disruption than the repulsion-only case.

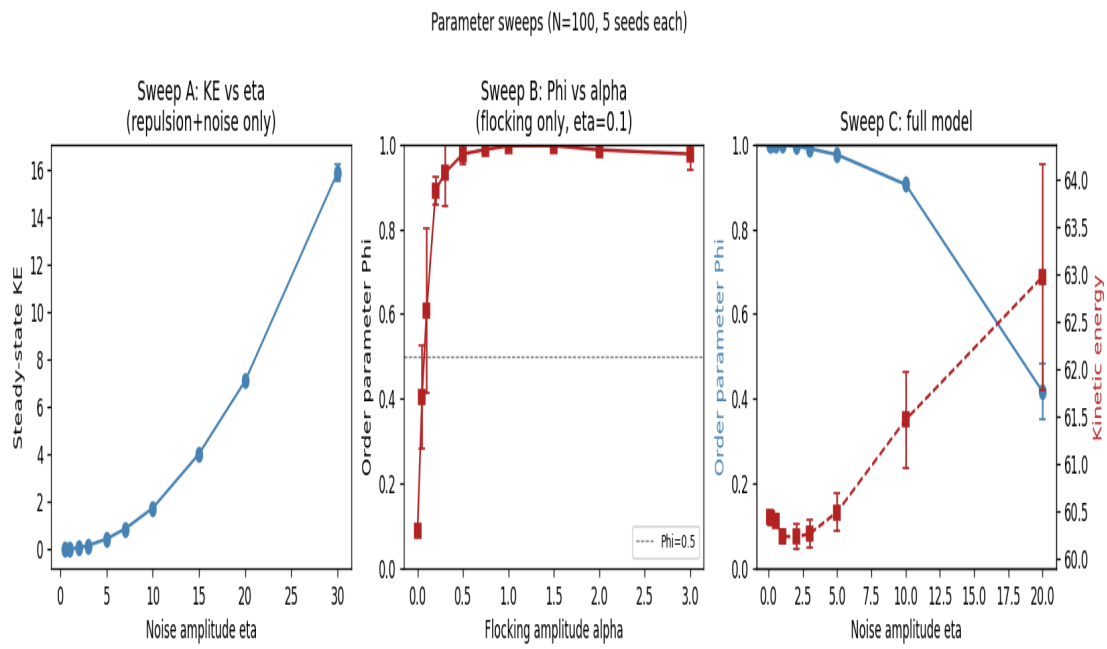

Figure 3. Flock formation sweep. Left panel: Φ vs. α (5 seeds, error bars = 1σ) showing sharp onset near $\alpha \sim 0.05$ with large variance in the transition region. Right panel: Φ vs. noise amplitude with full model active.

4.3 Nature of the Solid-to-Fluid Transition

In the repulsion-only system ($\alpha = 0$, $v_0 = 0$), kinetic energy rises with noise amplitude, suggesting a transition from a solid-like crystalline state to a fluid-like disordered state. To test whether this constitutes a true phase transition, I performed finite-size scaling across N = 25, 50, 100, and 200 (Fig. 4).

A true phase transition would produce KE/*N* curves that depend on *N*, with a critical point (susceptibility peak) that converges to a finite η_c as *N* increases. Instead, the KE/*N* curves are essentially identical for all four system sizes, and the susceptibility $\chi = N \cdot \mathrm{var}(KE/N)$ increases monotonically with η with no peak.

This indicates a smooth crossover rather than a true critical phenomenon. The physical picture is consistent with the high compactness of the system ($C = \pi N r_0^2 \sim 0.78$ for the parameters used): each agent is effectively caged by its neighbors and oscillates harmonically around a fixed lattice site. This produces KE proportional to η^2, independent of *N*—behavior characteristic of uncoupled harmonic oscillators, not a correlated system approaching criticality.

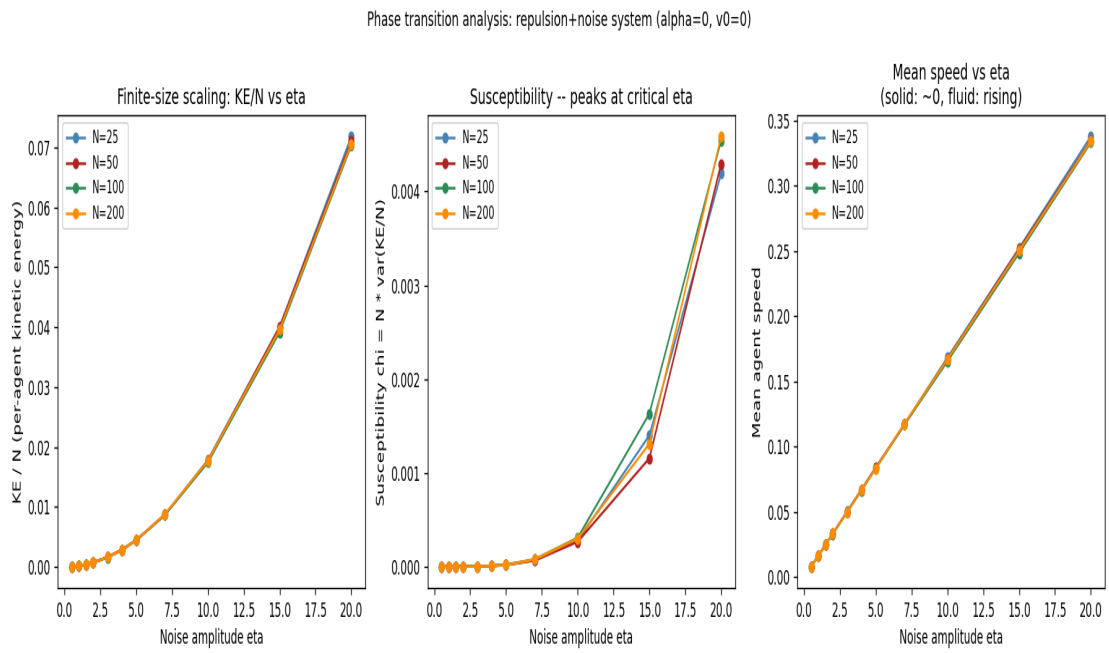

Figure 4. Finite-size scaling of the solid-to-fluid transition (N = 25, 50, 100, 200). Left: KE/N vs. noise amplitude η—all curves collapse onto a single line, indicating no N-dependence. Right: susceptibility $\chi = N \cdot \mathrm{var}(KE/N)$ rises monotonically with no peak, ruling out a true critical point.

4.4 Predator-Prey Dynamics

I extended the model with a predator agent that chases the prey center of mass via a strong alignment force ($\alpha_{pred} = 5$) and generates a long-range repulsive force on nearby prey ($r_{0,pred} = 0.1$). Prey parameters were set to the slow-walking regime ($v_0 = 0.02$, $\alpha = 1.0$, ramp = 0.1) to match the concert crowd context from Silverberg et al.

Flock coherence under pressure. Comparing flocking prey ($\alpha = 1$) versus non-flocking prey ($\alpha = 0$) across 10 random initializations shows a striking divergence. Flocking prey maintain $\Phi \sim 0.998$ throughout the simulation despite continuous predator pressure. Non-flocking prey scatter almost immediately, reaching $\Phi \sim 0.096$ in steady state (Fig. 5). Non-flocking agents maintain marginally more individual distance from the predator (0.127 vs. 0.112), but they lose all collective structure. The flock absorbs the disturbance while remaining coherent.

Evasion distance saturates. Sweeping predator aggression α_{pred} (which sets effective predator speed as $v_{eq,pred} = v_{0,pred} + \alpha_{pred}/\mu_{pred}$) reveals that the mean predator-to-nearest-prey distance drops from 0.24 with a passive predator to ~0.10 for $\alpha_{pred} \geq 1$ and then saturates—the collective repulsion response establishes a minimum buffer distance that persists regardless of predator aggression.

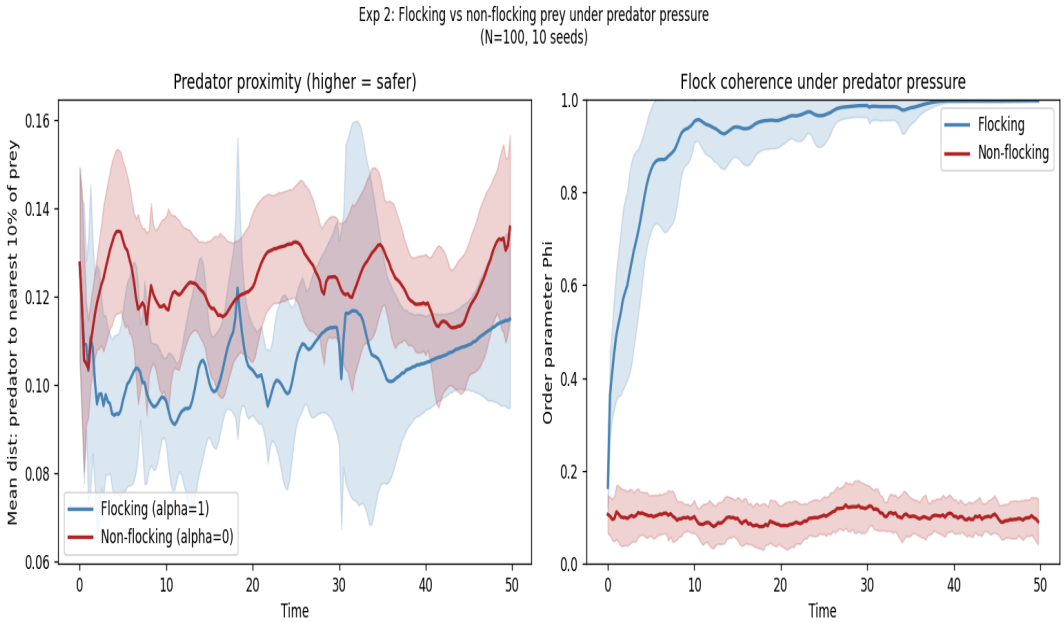

Figure 5. Flock coherence under predator pressure (10 seeds). Flocking prey (blue) maintain $\Phi \sim 0.998$ throughout. Non-flocking prey (red) scatter to $\Phi \sim 0.096$. Non-flocking agents are marginally farther from the predator (0.127 vs. 0.112) but lose all collective structure.

Flock geometry. Without a predator, the steady-state aspect ratio is AR = 2.61 and radius of gyration R_g = 0.215. With a predator, these shift modestly to AR = 2.76 and R_g = 0.274. Stronger flocking (larger α) produces substantially more elongated flocks: AR increases from 2.09 at α = 0.2 to 7.27 at α = 2.0. These highly elongated configurations resemble the arched, thinning flocks predicted by Charbonneau Exercise 6 (Fig. 6).

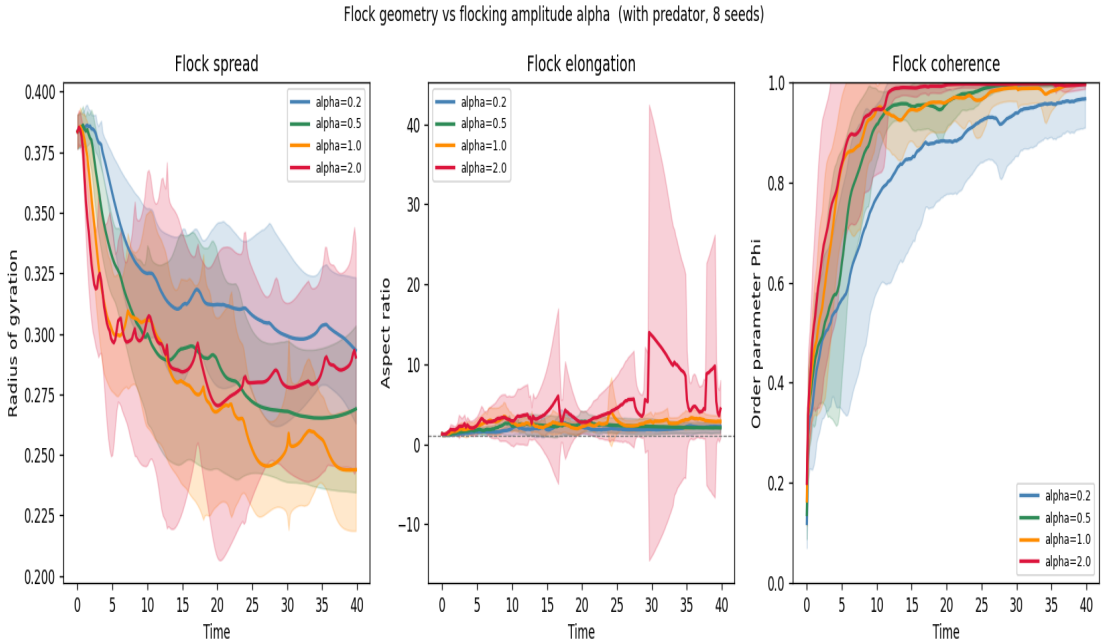

Figure 6. Flock geometry vs. flocking amplitude α under predator pressure (8 seeds). Aspect ratio rises sharply with α, reaching AR = 7.27 at α = 2.0. Coherence Φ remains near 1.0 in all cases.

Multiple predators. With 1 to 4 simultaneous predators, flock order parameter stays near 0.975–0.991—coherence is maintained across the entire range (Fig. 7). Aspect ratio rises substantially with predator count (AR = 2.83 with one predator, AR = 7.91 with three), while R_g increases modestly. Counterintuitively, the minimum predator-to-prey distance increases slightly as the number of predators grows (0.093 for one predator, 0.106 for three). The flock under multiple predators is more elongated but no more accessible to any individual predator.

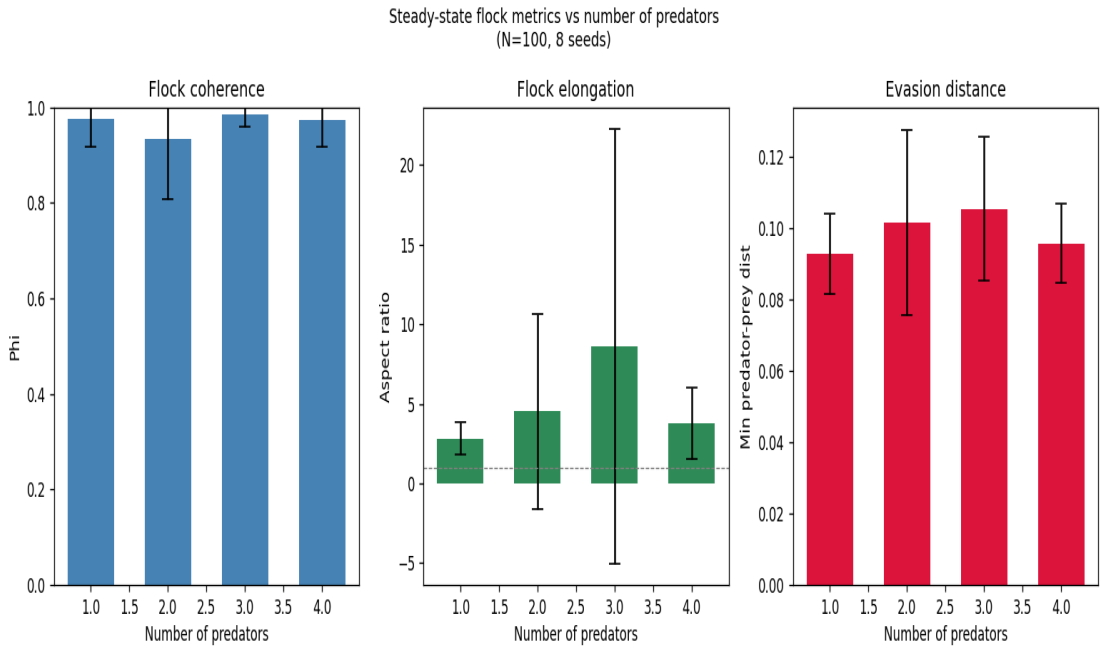

Figure 7. Steady-state flock metrics vs. number of predators (N = 100, 8 seeds). Coherence (Φ, left) stays near 1.0 regardless of predator count. Aspect ratio (center) rises from 2.83 to 7.91. Evasion distance (right) counterintuitively increases with more predators.

5. Discussion

The most striking result of the predator simulations is that flocking is not primarily a strategy for maximizing distance from a predator. Non-flocking agents individually maintain slightly greater separation from the predator, yet the flocking agents clearly have a more robust collective response: they remain coordinated, move in concert, and hold a consistent buffer distance. This distinction—coherence versus distance—may reflect something real about the function of biological flocking. A coherent flock can mount a coordinated escape response; scattered individuals cannot.

The increasing aspect ratio under multiple predators is interesting. When pressure arrives from multiple directions, the flock elongates rather than fragmenting. This could represent a geometric optimum: a thin stream is a harder target to surround than a compact cluster, and threading between predators may minimize total exposure. That this behavior emerges spontaneously from

the force equations, without any deliberate strategy encoded in the agents, is a good example of emergence.

The equilibrium speed result ($v_{eq} = v_0 + \alpha/\mu$) is an exact consequence of the force equations that Charbonneau does not explicitly note. It means a researcher using this model who sets $v_0 = 1$ and $\alpha = 1$ expecting agents to cruise at speed 1 will find them consistently at speed 1.1. For simulations where absolute speed matters (e.g., comparing flocking and non-flocking agents under identical predator pressure), this correction is necessary.

The phase transition result is perhaps the most physically interesting negative result. The expectation from analogies with equilibrium statistical mechanics—that a system with a solid-to-fluid transition should show a sharp critical point—does not hold here. The *N*-independence of KE/*N* and the absence of a susceptibility peak indicate that the agents in this high-compactness regime are not behaving cooperatively at the transition. Each agent vibrates independently. Whether a true phase transition could be found in a lower-compactness regime (where agents have more room to move) is an open question.

6. Conclusions

This study produced four main results:

1. Equilibrium speed: The cruise speed of an aligned flock is $v_{eq} = v_0 + \alpha/\mu$, exactly. This is a direct consequence of the force equations and must be accounted for when comparing simulations at different parameter values.

2. Phase transition: The solid-to-fluid transition in the repulsion-only system is a smooth crossover, not a true phase transition. Finite-size scaling shows no N-dependent critical point, consistent with independent harmonic oscillator behavior in a high-compactness system.

3. Flock coherence under predation: Flocking prey maintain near-perfect velocity alignment under predator pressure while non-flocking prey scatter. Evasion distance saturates at a minimum buffer value regardless of predator aggression.

4. Collective geometry: Flocks become more elongated under both stronger internal alignment and greater predator pressure. Multiple predators elongate the flock without degrading coherence; evasion distance counterintuitively improves.

Taken together, these results suggest that the primary function of the alignment force in this model—and possibly in biological flocking—is not to keep individuals far from threats, but to maintain coordinated collective response.

. References

Charbonneau, P. (2017). *Natural Complexity: A Modeling Handbook*. Princeton University Press.

Silverberg, J. L., Bierbaum, M., Sethna, J. P., and Cohen, I. (2013). Collective motion of humans in mosh and circle pits at heavy metal concerts. *Physical Review Letters*, 110, 228701.

. Appendix: Code

All simulation code is available at https://github.com/ninjahawk/Summer_Research.

File	Description
flocking.py	Core model: buffer zone, vectorized force function, run loop, metrics
analysis.py	Validation limiting cases and parameter sweeps
predator.py	Single-predator extension with 4 experiments
phase_transition.py	Finite-size scaling of solid-to-fluid transition
geometry.py	Radius of gyration and aspect ratio analysis
multi_predator.py	Multi-predator experiments
build_report.py	Generates this PDF using reportlab